Syllabus for Math 243: Calculus III (Section 001)

Spring 2025 (Jan 13 - May 16)

Instructor: Max Hill
Office: Physical Sciences Building 304
Office Hours: MWF 2-3pm (or other times by appointment).

Place and time: Watanabe 112 at 10:30-11:20am (MWF)

Textbook: *Calculus* by James Stewart (8th edition). ISBN-13: `978-1-305-26665-0`

Course prerequistes: a grade of C or better in Math 242 or Math 252A. If you don't meet the prerequisites, you'll need to get approval from the math department office.

Grading: Homework (30%), Midterm (35%), Final (35%). The following grade cutoffs will be used at the end of the semester to determine final grades:

D	D+	C-	C	C+	B-	B	B+	A-	A	A+
60%	67%	70%	73%	77%	80%	83%	87%	90%	93%	97%

I may change the above numbers, but if I do so it will be in your favor.

Homework: Homework will be due approximately weekly.

- You have one free 'no questions asked' homework extension.
- Your lowest homework score will be dropped at the end of the semester.
- You may collaborate with classmates on the homeworks. But if you do so, you must (1) make an effort write up your solutions on your own, using your own words, and (2) list the names of the people who you worked with.

Exams: There will be a midterm and a final exam. The final will be cumulative, but will emphasize material after the midterm. The exam dates are as follows:

Midterm exam .... Monday, March 24 (in class)
Final exam .... Monday, May 12, 9:45-11:45am

If you do better on the final than the midterm, then your grade on the final will replace your midterm grade.

Make-up policy: Make-up exams are allowed only in three types of circumstances: (1) in accordance with university policies, such as conflict with a religious observation, (2) conflicts with another university-related event, or (3) exceptional circumstances, such as a last-minute medical or family emergency with verification. In the first two cases, notice must be given to the instructor two weeks in advance.

Accommodations: Any student who feels s/he may need an accommodation based on the impact of a disability is invited to contact me privately. I would be happy to work with you, and the KOKUA Program (Office for Students with Disabilities) to ensure reasonable accommodations in my course. KOKUA can be reached at (808) 956-7511 or (808) 956-7612 (voice/text) in Room 013 of the Queen Lili'uokalani Center for Student Services.

Tentative course outline:

- **Weeks 1-3:** Parametric equations and Polar coordinates

  1. Section 8.1: Arc length
  2. Section 10.1: Curves defined by parametric equations.
  3. Section 10.2: Calculus with plane curves.
  4. Section 10.3 Polar coordinates
  5. Section 10.4: Areas and lengths in polar coordinates
  6. Section 10.5: Conic sections.

- **Weeks 4-6:** Vectors and the geometry of space.

  1. Section 12.1: Three-dimensional coordinate systems.
  2. Section 12.2: Vectors.
  3. Section 12.3: The dot product.
  4. Section 12.4: The cross product.
  5. Section 12.5: Equations of lines and planes.
  6. Section 12.6: Cylinders and quadric surfaces.

- **Weeks 7-9:** Vector functions.

  1. Section 13.1: Vector functions and space curves
  2. Section 13.2: Derivatives and integrals of vector functions.
  3. Section 13.3: Arc length and curvature.
  4. Section 13.4: Motion in space: velocity and acceleration.

- **Weeks 10-15:** Partial derivatives.

  1. Section 14.1: Functions of several variables.
  2. Section 14.2: Limits and continuity
  3. Section 14.3: Partial derivatives.
  4. Section 14.4 Tangent planes and linear approximations
  5. Section 14.5: The chain rule.
  6. Section 14.6: Directional derivatives and the gradient vector.
  7. Section 14.7: Maximum and minimum values.
  8. Discovery project p.1010: Quadratic approximation and critical points (Taylor's formula for two variables).
  9. Section 14.8: Lagrange multipliers.